

OPERATING SYSTEMS

Memory Management

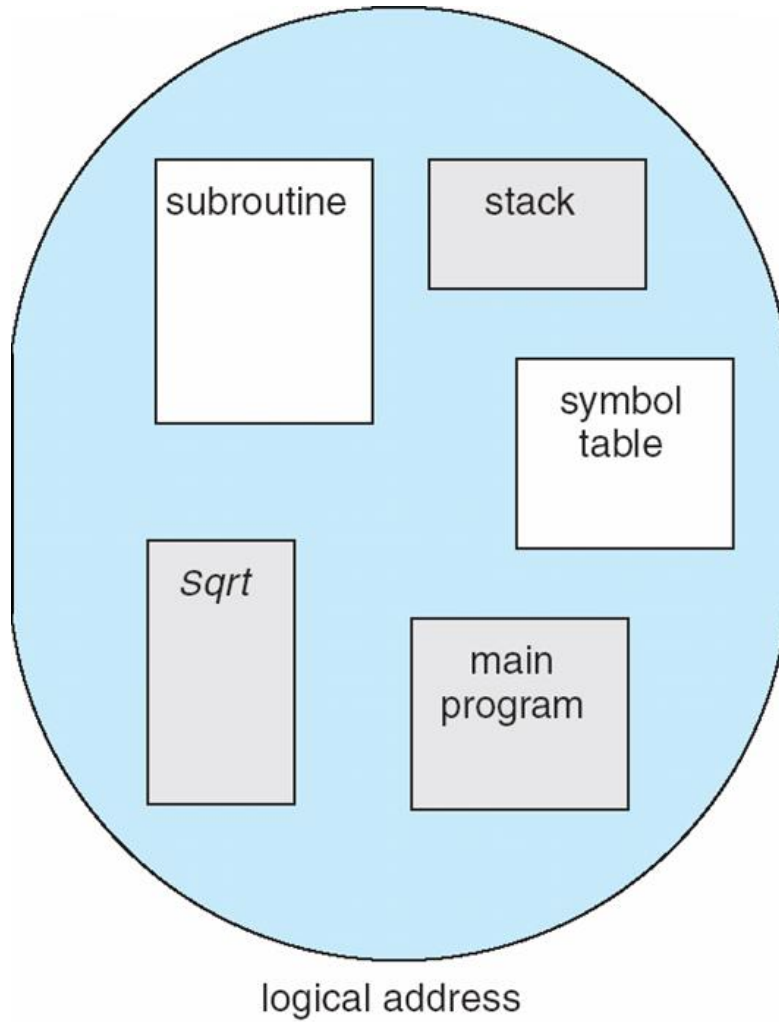
S.Thenmozhi

Department of Computer Applications

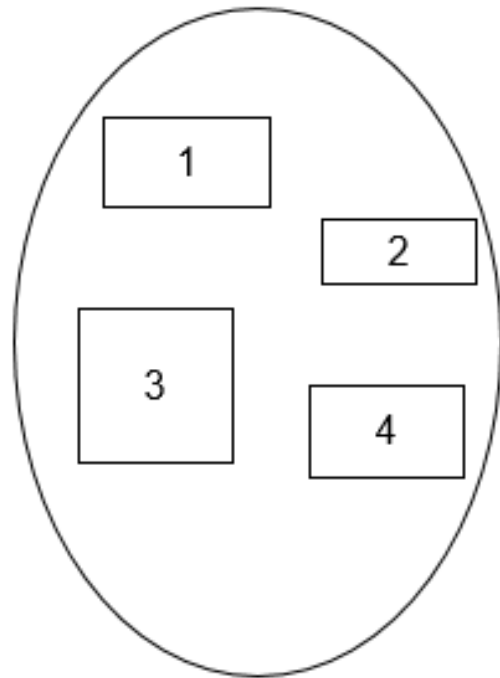
- A program is a collection of segments
- A segment is a logical unit such as:
 - main program
 - function
 - method
 - object
 - local variables, global variables
 - common block
 - stack
 - arrays

OPERATING SYSTEMS

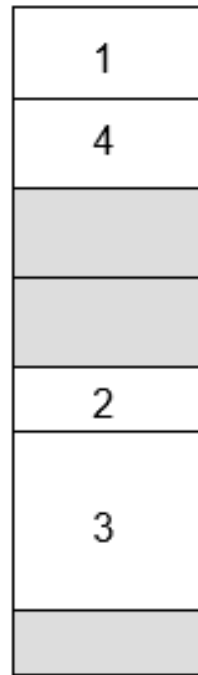
Segmentation



Logical View of Segmentation



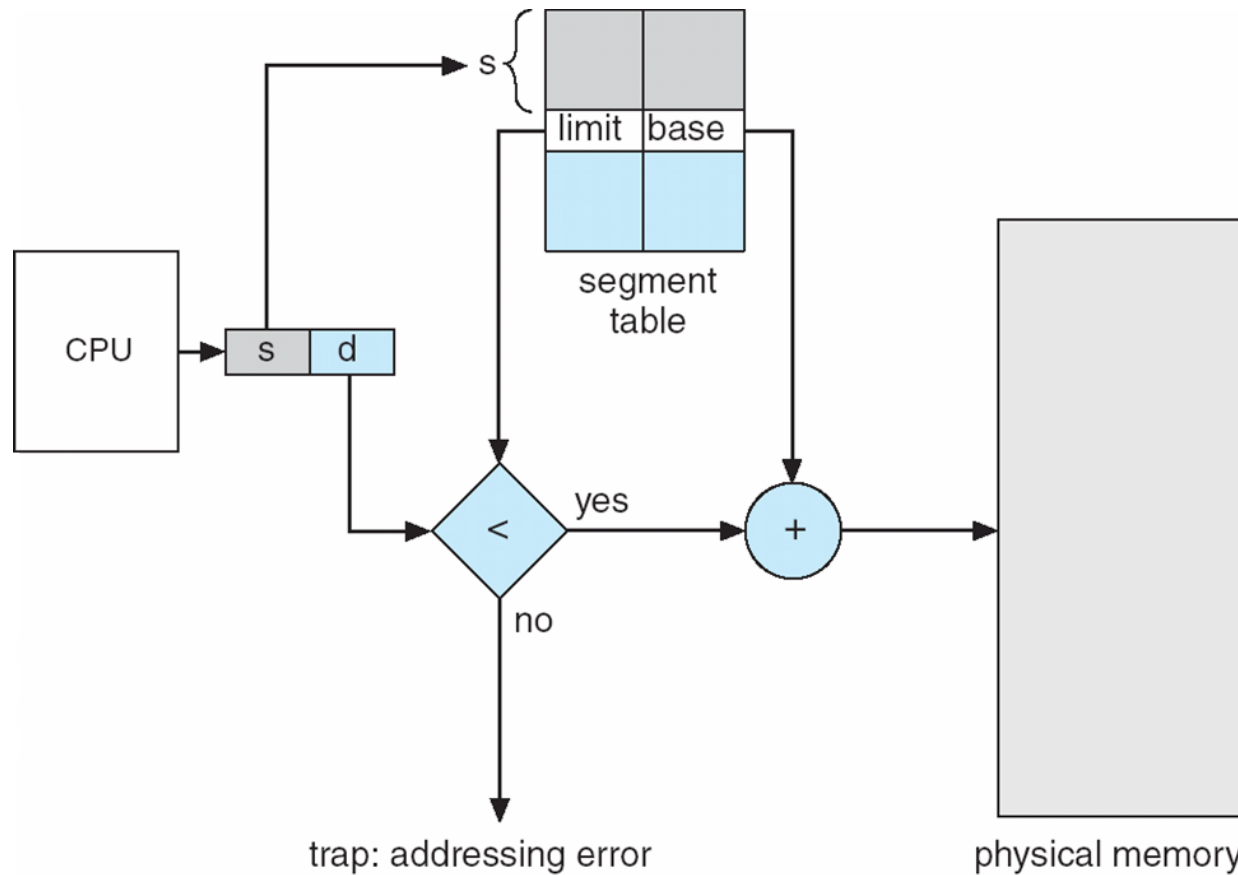
user space



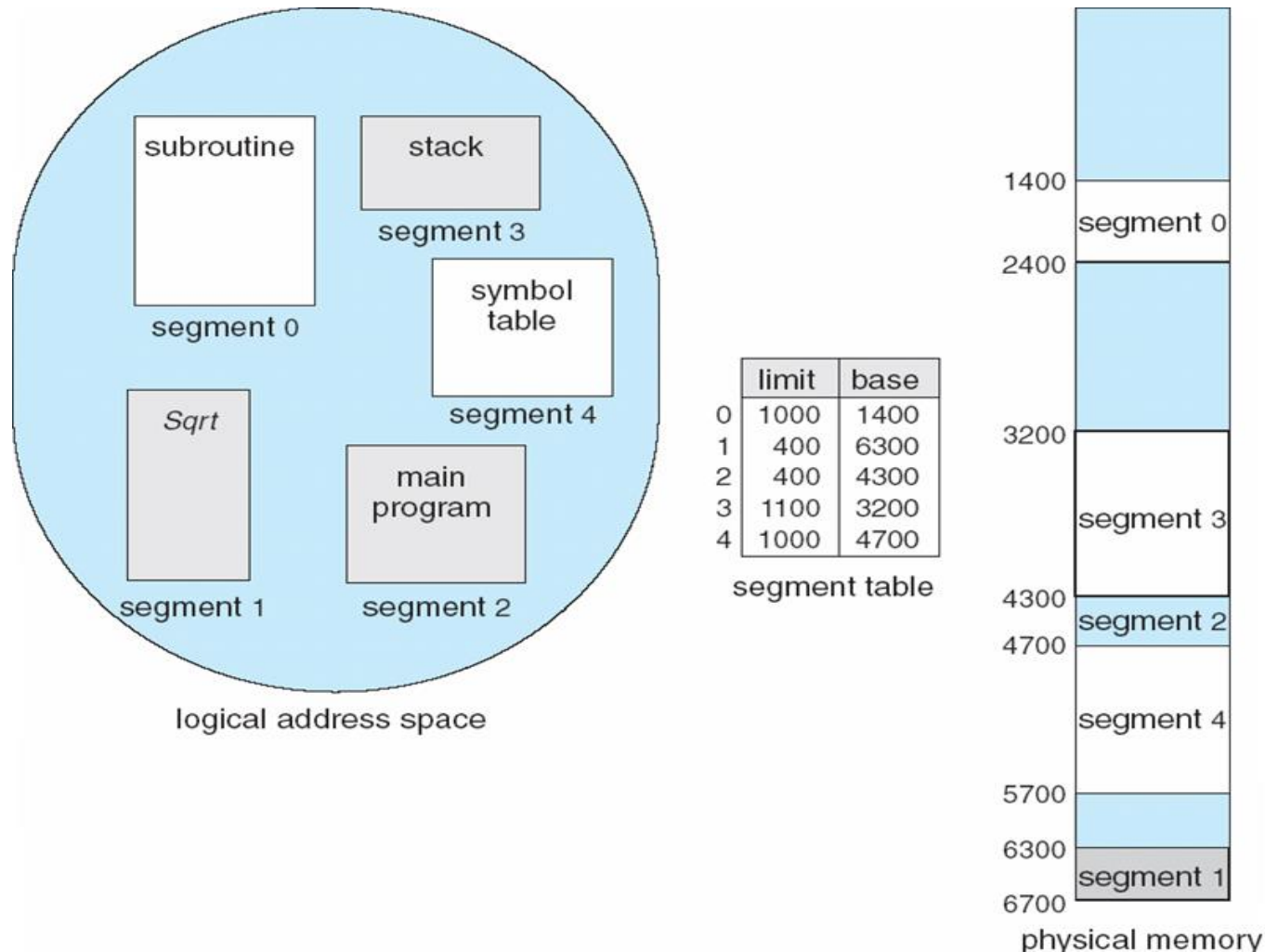
physical memory space

- Logical address consists of a two tuple:
 <segment-number, offset>,
- **Segment table** – maps two-dimensional physical addresses; each table entry has:
 - **base** – contains the starting physical address where the segments reside in memory
 - **limit** – specifies the length of the segment
- **Segment-table base register (STBR)** points to the segment table's location in memory
- **Segment-table length register (STLR)** indicates number of segments used by a program

Segmentation Hardware



Segmentation Example





THANK YOU

S. Thenmozhi

Department of Computer Applications

thenmozhis@pes.edu

+91 80 6666 3333 Extn 393